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ABSTRACT

This paper presents a study of the contact resistance between a metal M (M = Ni, Pt, and Au) and an array of *n*-type $Bi_2Te_{3-x}Se_x$ thermoelectric nanowires deposited through the electrodeposition process in the alumina membrane. Contact resistances between different metals and thermoelectric nanowires have been tested and characterized after optimization of the mechanical thinning and polishing process of the top part of the membrane. A low areal contact resistance of 87 μΩ cm^2 obtained with Au as the contact electrode is very encouraging for the development of thermoelectric modules based on nanowires in their membranes.

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I. INTRODUCTION

Bi_2Te_3 and its alloys are well known for their good thermoelectric (TE) performance close to room temperature.[1] Nanostructure engineering progress has further improved their performance, which has drawn much attention to thermoelectric devices.[2,3] Indeed, low dimensional materials are interesting due to efficiency enhancement by thermal conductivity reduction and their integration into microdevices.[4,5]

Although *n*- and *p*-type bismuth telluride nanowires (NWs) have been the subject of numerous studies,[6–30] only few approaches to build TE devices based on NW have been reported in the literature. Most of these are based on Si or SiGe NWs,[31–41] and few are based on Bi–Te or Bi–Sb–Te NWs.[42–45] The efficiency of TE devices depends on the dimensionless figure of merit $ZT = \alpha^2 T/\rho\lambda$ (with α, ρ, and λ being the Seebeck coefficient, electrical resistivity, and thermal conductivity, respectively) and its electrical contact properties.[46]

In the past, different approaches have been used to measure α and ρ and hence the power factor α^2/ρ. The simplest one involves the use of a two-point method, which consists of contacting both ends of a nanowire. For instance, conducting AFM has been used to measure the power factor of bismuth telluride single nanowires still embedded in their alumina template.[47] Nanoprobing has also been used to characterize the electrical resistivity of *n*- and *p*-type bismuth telluride nanowires by contacting with a W tip the nucleus generated by the overgrowth of nanowires from the template.[48] Another frequently used approach to measure these parameters is the so-called four-point method in which a single nanowire is electrically contacted to four points. Several types of metallic contacts have been studied, such as Ni,[49] Al,[50] Cr/Au,[51] or Au/W.[52] Unlike the two-point method, the four-point method eliminates measurement errors due to the probe resistance, the spreading resistance under each probe, and the contact resistance between the metal probe and the measured material.

There are many methods for contact resistance measurement, but only a few are actually employed in the nanomaterial field. Typically, the Transfer-Length Method (TLM),[53] four probe method,[54] and Y-function method (YFM)[55] are used for extracting the contact resistance in nanomaterials. Adapted TLM[56] and contact end resistance measurement methods[57] can also be suitable for metal contacts

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to low dimensional materials. In general, these methods require the integration of NWs into specific microchips, which can be highly tedious. In addition, it is difficult to achieve a reliable measurement to determine the intrinsic properties of contact resistance and distinguish them from external effects (adsorption of a contamination layer and NW oxidation). It is currently observed that the electrical contact resistance study of individual bismuth telluride NWs showed non-Ohmic behavior[58] because of oxygen contamination during preparation processes.

Our objective in this study is to keep the NWs within the alumina matrix in such a way that NWs are not exposed to oxidizing atmospheres that can modify NWs and contacts properties. In addition, it is known that four-point measurement preparation induces possible mechanical problems, which can lead to NW breaking.

Because of the thermoelectrical NW size that is close to 60 nm, contact resistances strongly depend on the nanojunctions between thermoelectrical materials and metals. The size of these nanojunctions will strongly depend on the capacities of the metals deposited to reduce the asperities at the interfacial contact. When two rough surfaces are connected together, the contact consists of patches of size that can go down to the nanoscale. These junctions at a nanometer scale often exhibit electrical and mechanical properties that diverge from bulk properties.[59] Various mechanisms such as quantum tunneling,[60–62] Sharvin contact,[63] and Holm contact[64] depending on the size of contacting junctions and the mean free path of electrons are used to explain transport properties. In addition, the nanojunction between rough surfaces can be increased up to a microscale scale under pressure. The size of the patches is strongly dependent on the elastic properties of both materials. The inevitable presence of resistive surface films such as oxide layers also contributes to the interfacial resistance. Under sufficient pressure, surface asperities can penetrate the oxide layer, thus increasing metal-to-metal contact patches resulting in a relative low resistance.

In this paper, we present a first study of the contact resistance between *n*-type TE bismuth telluride NWs and different metals such as Ni, Pt, and Au. A mechanical thinning and optimized polishing process of the membrane is first presented. The diffusion of deposited metals into the TE material is then investigated by energy dispersive x-ray (EDX) analysis before contact resistance measurements of large NW area. The configuration measurement used in this study is exactly the one that will be used to develop thermoelectric microgenerators from NWs in their alumina matrix. Indeed, *n*-type and *p*-type NWs assemblies must be electrically connected by a metal with a contact resistance that must be as low as possible to limit the internal resistance of the microdevice. We believe that finding a technique estimating the contact resistance close to microdevice configurations is very useful for the development of thermoelectric microdevices based on NWs. The low value of areal contact resistance obtained with gold metal is very interesting for the development of autonomous systems based on micro-thermal harvesting.

II. EXPERIMENTAL DETAILS

A. Nanowires growth by electrochemical process and metal deposition

Appropriate $Bi_2Te_{3-x}Se_x$ nanowires leading to *n*-type semiconducting elements have been developed using pulsed electrodeposition from aqueous solution at room temperature. The electrolyte used is composed of $HClO_4$ 1M solution containing 10 mM Bi_2O_3, 10.3 mM TeO_2, and 1.1 mM SeO_2. The matrix consisted of nanoporous anodic aluminum oxide membranes fabricated by a two-step anodization process in 0.5M oxalic acid at 40 V to ensure highly ordered pore arrays.[65] The template thickness and pore diameter are 60 μm and 60 nm, respectively. Pulsed electrodeposition conditions, microstructure, composition, and TE characterization techniques of electrodeposited nanowires have been detailed in previous works.[66,67]

EDX analysis of the *n*-type nanowire revealed a composition close to $Bi_{1.45}Te_{2.85}Se_{0.7}$. Electron diffraction patterns realized on several isolated nanowires are characteristic of nearly single-crystalline NWs in accordance with XRD diffractograms, which show a strong texture along the [1 1 0] direction. Composition homogeneity along the length of the NW was verified on a single NW using TEM[48] and on NW bundles obtained using EDX after dissolution of the anodic aluminum oxide (AAO) membrane. A composition variation along the NW axis lower than 10% was observed.

Values obtained for α, ρ, and λ are -78 μV/K,[66] 0.29 mΩ cm,[48] and 1.25 W/m K[67] at 300 K, respectively.

Different metals were deposited on the top part of the AAO membrane using the sputtering process. Au metallization necessary for the electrochemical process was kept on the bottom part of the AAO membrane. Three or four pads of 1 mm diameter and 200 nm thickness were deposited after surface preparation thinning and etching, as described below. Films were deposited at room temperature using a dc magnetron sputtering device. Ni Pt and Au commercial targets of 50 mm diameter were used. The pressure before deposition was lower than $5 \cdot 10^{-6}$ mbar. A 10 min target pre-sputtering was used to eliminate surface contamination prior to plasma deposition. Deposition was realized with the DC power fixed to 100, 150, and 200 ± 4 W for Au, Ni, and Pt targets, respectively, and with 8 × 10^{-3} mbar argon pressure. The reactive ion etching (RIE) conditions were fixed to 240 V/1 A with $3 \cdot 10^{-4}$ mbar Ar pressure.

B. Membrane surface preparation

1. Mechanical thinning and optimized polishing process

Differences in the nanowires growth rate and unfilled pores lead to membranes partially filled with a non-sharp growth front. It has been shown that pulsing the potential between a deposition potential and the other one leads to a more abrupt growth front.[68] The pulsed regime also leads to a more homogeneous chemical composition along the wire axis and improves crystallinity. In our case, the pulsed regime has ensured a better filling ratio. Surface mechanical thinning and polishing are still necessary to overcome the 20 μm of unfilled pores, as shown in Fig. 1(a).

Membrane brittleness makes the polishing process a critical step due to porosity, thickness, and stresses during electrochemical deposition. Many membranes break during polishing if the process is not optimized. In this context, we developed a special sample holder and a constant pressing process that leads to a planar and homogeneous polishing and avoids membranes loss [Figs. 1(b) and 1(c)]. Different grain size abrasives and polishing durations were also studied to optimize the surface quality finish [Figs. 1(e) and 1(f)]

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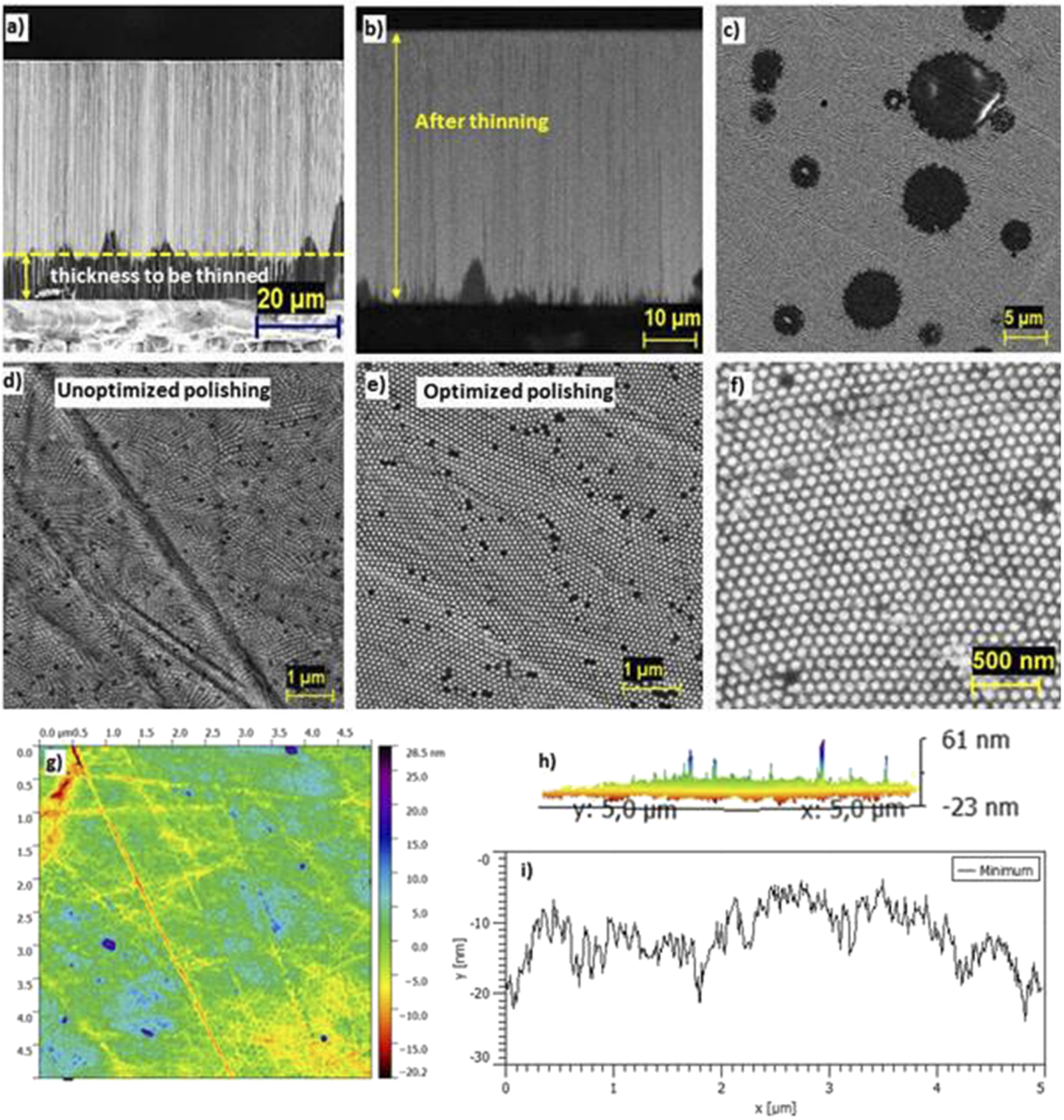


FIG. 1. [(a) and (b)] SEM images of the membranes in the cross section before and after thinning and polishing. (c)–(f) SEM images of surfaces after thinning and polishing. (g) AFM topography on a polished surface (5 × 5 μm^2). (h) 3D topography. (i) Depth profile of polishing marks (in nm).

and to avoid deep and irregular polishing marks [Fig. 1(d)]. Only mechanical polishing is applied on the face with overgrown material (the back side corresponding to the bottom of the membrane being still covered by gold used as the contact electrode for electrochemical deposition of the material). It starts with a grit paper of 1200, down to 4000, and finishing is done with diamond paste of 1 μm. No chemical or electrochemical polishing was used at this step. As evidenced by the SEM pictures [Figs. 1(e) and 1(f)], no SiC grain incorporated in the alumina filled surface after polishing can be observed.

AFM topographic analysis was performed to estimate the depth of the polishing marks, as shown in Figs. 1(g) and 1(h). Analysis on different spots of 25 μm^2 high quality surface shows a good planarity distribution with marks of depth lower than 10 nm [Fig. 1(i)]. Under

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these conditions, we estimate that the percentage of unfilled pores is lower than 10%.

2. Impact of surface RIE (reactive ion etching)

The second crucial step of samples surface preparation is ion etching before contact metal deposition. It removes the oxide layer on the polished surface and improves the metallic contact at the nanowire tips.

Areal contact resistance ρ_c calculated with Eqs. (1)-(4) was studied for different etching durations to optimize oxide layer removal. A constant metal thickness of 200 nm was chosen for this study. Samples with no etching revealed high ρ_c (6.7 mΩ cm^2), which confirms etching as a key parameter.

Best results are obtained for 30 and 90 s leading to ρ_c lower than 500 $\mu\Omega$ cm^2. Very high etching durations (2–7 min) induced higher ρ_c (up to 1 mΩ cm^2). These high ρ_c values are explained by sample surface heating during etching, leading to a local composition modification of $Bi_2Te_{3-x}Se_x$. Additional depositions with different contact metals were then processed on both types of etched samples (30 and 90 s etching duration). The best areal contact resistance of 200 $\mu\Omega$ cm^2 was obtained for 90 s etching time. This duration is adopted for the rest of this study.

III. RESULTS

A. Study of metal diffusion

In order to complete our study, we focused on samples surface analysis after metal deposition. Figure 2 shows captions of AFM surface analysis (25 μm^2) after each metal deposition (Au, Ni, and Pt). Topography results show that maximum depth of polishing marks is around 20 nm. The contact deposition process used for NWs is of high quality, and no metal delamination is observed as for thermoelectric films.[69]

Diffusion of deposited metals into the TE material was also investigated. Characterization process issues complicated the study of metal diffusions at the nanometric scale. It was still possible to establish a micrometric investigation of Ni diffusion described by the concentration evolution profile [Figs. 3(a) and 3(b)].

Further study was conducted on Ni diffusion into samples. The polished TE material with 200 nm thick Ni deposits was analyzed on the cross section at the interface by SEM-EDX. Figure 3(a) shows the microstructure of the contact region (cross section) between the Ni layer and TE nanowires. A sharp separation and no additional layer are observed between the Ni and NWs illustrating a low Ni diffusion into the sample and no formation of the intermediate component. These conclusions are confirmed by EDX spectra obtained on different analysis points on both sides of the contact region, as shown in Fig. 3(b). For points D–G, the electronic analysis beam was centered on an emerging Bi–Te–Se NW after fracture. According to Monte Carlo simulation, the interaction volume of the analysis electron beam is estimated to 100 nm and can be represented by uncertainty bars on the position x. In this condition, the small Ni concentration found at the +5 nm could be due to the analysis rather than to a real diffusion of Ni in the Bi–Te–Se NW. Despite these difficulties in analysis, it seems that the Ni peak appears at 5 nm from the contact region and disappears 20 nm further.

Figure 4 shows that the diffusion is increased for Pt and Au. The concentrations in Pt and Au become close to zero at a distance of 0.5 and 1 μm, respectively, from the interface [Fig. 4(b)].

B. Areal contact resistance ρ_c measurements

The electrical contact resistance was measured with a scanning voltage probe setup at room temperature and at atmospheric pressure under air. The voltage drop along a TE NW is measured as a small current passes through the element. The diameter of the

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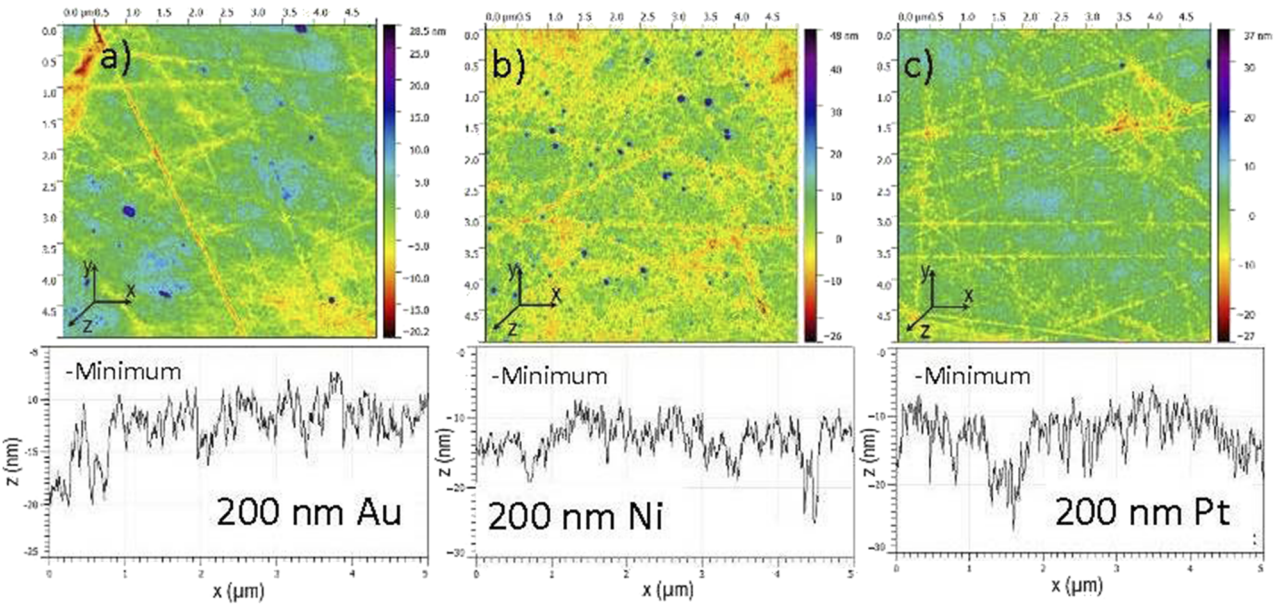


FIG. 2. AFM topography of the different surfaces after deposition of Au (a), Ni (b), and Pt (c).

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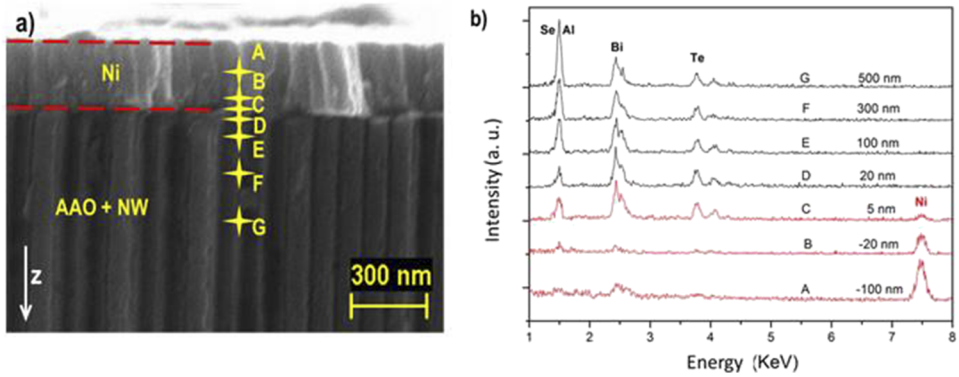


FIG. 3. (a) Cross section image of the Ni/NWs interface. EDX analysis points are visible in yellow. (b) EDX spectra at the analysis points.

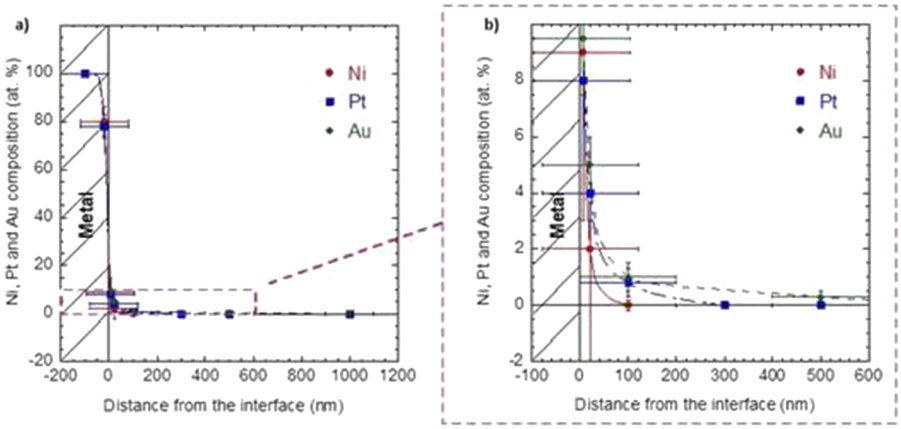


FIG. 4. (a) Variation of Ni, Pt, and Au concentrations as a function of the distance from the interface. (b) Zoomed-in view of a part of Fig. 4(a) delimited by red dashes.

circular pad for resistance measurement is 1 mm, and a current varying from 0 to 10 mA is applied. The measurement is made with the current flowing in the forward direction (from the top to the bottom of the NWs), and then, a second measurement is made with the current in the reverse direction in order to subtract any thermal electromotive force. The temperature rise induced by Joule's effect is estimated to be less than 0.5 K. Figure 5 shows the experimental setup for the contact resistance measurements.

An xy-table allows horizontal movement of the membrane. Adjustment screws and a manipulator arm connected to the measuring tips are used to achieve an extremely precise contact between tips and pads. The force exerted by the point of our measurement resistance setup was determined using a commercial accuracy balance. The maximum of the force measured is 0.3 N. The maximum stress applied to the contact is 1.5 MPa according to a point surface of 0.2 mm^2.

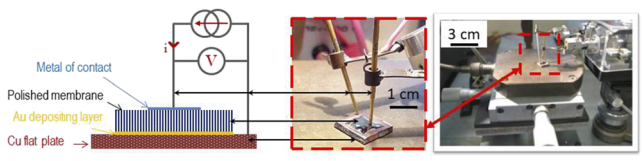


FIG. 5. Experimental setup for contact resistance measurements.

During the measurement of our samples, by increasing the force of the point on the contact, the resistance passes sharply from insulating to conducting. Once contact has been established, no resistance variation vs the pressure is observed. We concluded that the pressure has no influence on the resistance measurements.

The contact resistance can be calculated according to the following equation:

$$R_{measured} = \frac{V}{i} = R_{NW} + R_{set_up} + R_c^{sup} + R_c^{inf}, \quad (1)$$

with R_{NW} being the NW electrical resistance and R_{set_up} being the setup resistance. R_c^{sup} and R_c^{inf} correspond to the electrical contact resistance of the metal/NW heads and metal/NW feet. The resistance of different components and connections (two copper wires, two copper wire/clamping pieces, two clamping of the retractable wire, and two retractable tips of 1.5 cm length and 0.8 mm diameter) corresponding to R_{setup} is 1 mΩ. According to the electrochemical process for Au metallization of the bottom part of the AAO membrane, it was assumed that $R_c^{inf} \ll R_c^{sup}$. Contact resistance R_c and areal contact resistance ρ_c are calculated by the following equations:

$$R_c = R_c^{sup} = R_{measured} - R_{NW}, \quad (2)$$

$$\rho_c = R_c \cdot S_{NW}, \quad (3)$$

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with S_{NW} being the contact area of connected NWs, which can be obtained by the following equation:

$$S_{NW} = d \cdot S_{pad}, \quad (4)$$

where d is the areal NWs density and S_{pad} is the contact area of the metallic circular pad.

With the AAO prepared for this study and $S_{pad} = 0.785\,mm^2$, S_{NW} is close to $2.35 \times 10^{-3}\,cm^2$.

In previous studies,[48] the electrical resistivity is determined as 0.29 mΩ cm. The NW density and dimension lead to an R_{NW} of 0.56×10^{-3} Ω. After thinning, pores may remain unfilled, and as mentioned before, we estimated that the percentage of unfilled pores is less than 10% of the total pores inside a contact pad. The increase in the NW resistance induced by these unfilled pores is lower than 11%. With the NW resistance R_{NW} being about 1.5% of the contact resistance R_c, these errors have no influence on the areal contact resistivity ρ_c.

Table I summarizes the areal contact resistances of the studied metals (Pt, Ni, and Au) deposited by sputtering (200 nm) with 90 s etching time. The values reported here are averages of measurements realized from three or four contact pads deposited on the sample at different locations. The variations of the measurements are lower than 10% and are in agreement with the estimated unfilled pores percentage. Results show the lowest areal contact resistance for Au with a value of 87 $\mu\Omega\,cm^2$. This value can be reduced by 10% if 10% of the pores are unfilled. Pt and Ni also show low values in the range of several hundreds of $\mu\Omega\,cm^2$.

In the future, we will tend to use the photolithography procedure in order to reduce the pad size to the scale of a few μm^2. Statistical study should be realized using a higher number of micrometric pads than what we have achieved so far. The density of unfilled pores should also be reduced by decreasing the pad size.

We also studied the impact of an adhesion layer on the electrical resistance. A deposition of 10 nm of W shows an increase in contact resistance in comparison with samples metallized directly. Adding to its high diffusion behavior, we considered that the W layer was not necessary and even harmful to obtaining low contact resistance.

The low value obtained for Au metal contacts is crucial for performance improvement of TE modules. It is comparable to those obtained in the literature on Bi_2Te_3 TE elements based on bulk[70,71] and thin films[72] with Ni or Bi–Sn as the contact metal between Bi_2Te_3 and electrodes.

The lowest contact resistances were reported by Gupta *et al.* They obtained values of about 5 $\mu\Omega\,cm^2$ on Bi_2Te_3/Ni and Sb_2Te_3/Ni interfaces.[73] This value was subsequently improved to a value lower than 0.1 $\mu\Omega\,cm^2$ on thin layers of Bi_2Te_3, with Ni and Co as the contact metals.[74] The high contact resistance of the NW/Ni interfaces could be due to a less pronounced diffusion in the BiTeSe NW compound than for Au and Pt. The effects of recombination between oxygen from the AAO membrane and Ni element leading to nickel oxide formation are also not to be neglected to explain these high values compared to those of the literature.

TABLE I. Areal contact resistances ρ_c obtained for the different metals studied. R_{NW} has been determined using the electrical resistivity measured in Ref. 48 and NW dimensions.

Metal	Ni	Pt	Au
$R_{measured}$ (Ω)	0.1	0.08	0.038
R_{NW} (Ω)	0.56×10^{-3}	0.56×10^{-3}	0.56×10^{-3}
R_c (Ω)	0.099	0.079	0.037
ρ_c ($\mu\Omega\,cm^2$)	232	185	87

IV. CONCLUSION

Different metals such as Ni, Pt, and Au have been studied for the optimization of the contact resistance between $Bi_2Te_{3-x}Se_x$ NW in AAO membrane and electrodes. EDX analysis shows that the highest diffusion is for Au, whereas the diffusion depth of Ni is lower than 15 nm. After focusing on surface preparation of the AAO membrane with NW still embedded in it and the etching conditions, a low areal contact resistance of 87 $\mu\Omega\,cm^2$ has been obtained with Au metal contact. Such results are very encouraging for the development of thermoelectric modules based on nanowires in their membranes.

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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